

# While Loops in Python

## Unknown Repetition, Validation and Menus

Programming Fundamentals Team

SETU

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# Outline

**1. Learning outcomes**

**2. From selection to iteration**

**3. Basic while loops**

**4. Counting loops**

**5. Accumulators and counters**

**6. Validation loops**

**7. Sentinel loops**

**8. Menu loops**

**9. Infinite loops**

**10. Bringing it together**

**11. Summary**

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# 1.1 By the end of this lecture

You should be able to:

- Explain when a `while` loop is useful
- Write a simple `while` loop in Python
- Use counters and accumulators
- Write validation loops
- Write sentinel-controlled loops
- Create a simple menu loop
- Identify and avoid infinite loops

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## 2.1 You already know selection

```
age = int(input("Enter age: "))

if age >= 18:
    print("Adult")
else:
    print("Minor")
```

Selection makes a decision once.

## 2.2 What if we need to keep deciding?

Example:

- Keep asking for a password until it is correct.
- Keep asking for a mark until it is between 0 and 100.
- Keep displaying a menu until the user chooses Exit.

**A loop lets us repeat a block of code.**

## 2.3 Selection versus while loop

```
if condition:  
    print("Runs once if condition is true")
```

```
while condition:  
    print("Runs again and again while condition is true")
```

The condition is checked before each repetition.

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## 3.1 Syntax

```
while condition:  
    statement_1  
    statement_2
```

Important:

- The colon : starts the loop body.
- Indentation shows which statements belong inside the loop.

## 3.2 First example

```
count = 1

while count <= 5:
    print("Hello")
    count += 1
```

This prints Hello five times.

## 3.3 Anatomy of the loop

```
count = 1          # setup

while count <= 5:  # test
    print(count)   # repeated action
    count += 1     # update
```

**Setup, test, action, update.**

## 3.4 Trace table

```
count = 1

while count <= 3:
    print(count)
    count += 1
```

| count | condition | output |
|-------|-----------|--------|
| 1     | True      | 1      |
| 2     | True      | 2      |
| 3     | True      | 3      |

## 3.4 Trace table

|   |       |      |
|---|-------|------|
| 4 | False | Stop |
|---|-------|------|

## 3.5 Activity: predict the output

```
x = 3

while x <= 9:
    print(x)
    x += 2
```

Questions:

- What is the first value printed?
- What is the last value printed?
- Why does the loop stop?

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## 4.1 Count from 1 to 10

```
i = 1

while i <= 10:
    print(i)
    i += 1
```

## 4.2 Count backwards

```
i = 10

while i >= 1:
    print(i)
    i -= 1
```

## 4.3 Count by twos

```
i = 0

while i <= 20:
    print(i)
    i += 2
```

## 4.4 Quick challenge

Write a while loop that prints:

```
5
10
15
20
25
30
```

## 4.5 Quick challenge Solution

```
x = 5

while x <= 30:
    print(x)
    x += 5
```

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## 5.1 Counter pattern

A counter keeps track of how many times something happens.

```
count = 0  
count += 1
```

Examples:

- number of valid marks entered
- number of failed login attempts
- number of products sold

## 5.2 Accumulator pattern

An accumulator builds up a total.

```
total = 0  
total += value
```

Examples:

- total marks
- total cost
- total distance

## 5.3 Sum numbers 1 to 5

```
total = 0
num = 1

while num <= 5:
    total += num
    num += 1

print(total)
```

Output:

```
15
```

# 5.4 Trace the accumulator

| num | total before | total after |
|-----|--------------|-------------|
| 1   | 0            | 1           |
| 2   | 1            | 3           |
| 3   | 3            | 6           |
| 4   | 6            | 10          |
| 5   | 10           | 15          |

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## 6.1 Why validation loops?

User input cannot always be trusted.

Examples:

- A mark must be between 0 and 100.
- A menu choice must be one of the options.
- A password must match.

**A validation loop repeats until the input is acceptable.**

## 6.2 Validating a mark

```
mark = int(input("Enter mark: "))

while mark < 0 or mark > 100:
    print("Invalid mark")
    mark = int(input("Enter mark: "))

print("Accepted")
```

This uses `or`, which students have already met.

## 6.3 Validating a positive number

```
number = int(input("Enter a positive number: "))

while number <= 0:
    print("That is not positive")
    number = int(input("Try again: "))

print("Thank you")
```

## 6.4 Password example

```
password = input("Password: ")

while password != "python":
    password = input("Incorrect. Try again: ")

print("Access granted")
```

## 6.5 Activity: fix the validation

This code accepts invalid marks. Why?

```
mark = int(input("Mark: "))

while mark < 0 and mark > 100:
    print("Invalid")
    mark = int(input("Mark: "))
```

Hint: think about and versus or.

## 6.6 Activity: Solution

This code accepts invalid marks. Why?

```
mark = int(input("Mark: "))

while mark < 0 or mark > 100:
    print("Invalid")
    mark = int(input("Mark: "))
```

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## 7.1 What is a sentinel?

A sentinel is a special value that means stop.

Examples:

- -1 to stop entering numbers
- quit to stop entering commands
- 0 to stop entering product codes

## 7.2 Print numbers until -1

```
number = int(input("Number (-1 to stop): "))

while number != -1:
    print(number)
    number = int(input("Number (-1 to stop): "))
```

## 7.3 Total numbers until -1

```
total = 0
number = int(input("Number (-1 to stop): "))

while number != -1:
    total += number
    number = int(input("Number (-1 to stop): "))

print("Total:", total)
```

## 7.4 Count numbers until -1

```
count = 0
number = int(input("Number (-1 to stop): "))

while number != -1:
    count += 1
    number = int(input("Number (-1 to stop): "))

print("You entered", count, "numbers")
```

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## 8.1 Why menu loops?

Many programs keep running until the user chooses to exit.

Examples:

- ATM menu
- shop system
- student record system
- game menu

## 8.2 Basic menu loop

```
choice = ""

while choice != "4":
    print("1. Add")
    print("2. Search")
    print("3. Report")
    print("4. Exit")

    choice = input("Choice: ")
```

## 8.3 Menu with selection

```
choice = ""

while choice != "4":
    print("1. Add")
    print("2. Search")
    print("3. Report")
    print("4. Exit")

    choice = input("Choice: ")

    if choice == "1":
        print("Add selected")
    elif choice == "2":
        print("Search selected")
```

## 8.3 Menu with selection

```
elif choice == "3":  
    print("Report selected")  
elif choice != "4":  
    print("Invalid choice")
```

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## 9.1 Common cause

```
i = 1

while i <= 5:
    print(i)
```

The update is missing.

*i* never changes, so the condition never becomes false.

## 9.2 Intentional infinite loop

```
while True:
    command = input("Enter command: ")

    if command == "quit":
        break

    print("You entered", command)
```

It is better to write the loop condition so that the loop naturally stops, rather than using break

## 9.3 Better without break

```
command = input("Enter command: ")

while command != "quit":
    print("You entered", command)
    command = input("Enter command: ")

print("Quitting")
```

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## 10.1 Mini-program: average valid marks

```
total = 0
count = 0

while count < 5:
    mark = int(input("Enter mark: "))

    while mark < 0 or mark > 100:
        mark = int(input("Invalid. Try again: "))

    total += mark
    count += 1

average = total / count
print("Average:", average)
```

## 10.2 What does this combine?

- Input and output
- Selection-style conditions
- `or`
- Nested `while` loop
- Counter
- Accumulator
- Average calculation

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# 11.1 Today we learned

- `while` loops repeat while a condition is true.
- Counters count events.
- Accumulators build totals.
- Validation loops keep asking until input is valid.
- Sentinel loops stop when a special value is entered.
- Menu loops keep programs running.
- Infinite loops often happen when the update is missing.

## 11.2 Practice exercises

1. Print your name 10 times.
2. Count from 1 to 20.
3. Count backwards from 20 to 1.
4. Print even numbers from 2 to 50.
5. Calculate the sum of numbers from 1 to 100.
6. Ask repeatedly for a positive number.
7. Ask for marks until -1 is entered, then print the average.
8. Create a menu that repeats until the user chooses Exit.

# Thanks for Watching - Any questions?

